**Quiz 4 - DSC 10, Spring 2025**

Full Name:

PID:

Quiz Time: ◯ 3PM ◯ 3:30PM ◯ 4PM ◯ 4:30PM

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**Instructions:**

- This quiz consists of 3 questions. You have a total of 20 minutes to complete it.
- Please write **clearly** in the provided answer boxes; we will not grade work that appears elsewhere. Completely fill in bubbles and square boxes; if we cannot tell which option(s) you selected, you may lose points.

  ◯ A bubble means that you should only **select one choice**.

  ☐ A square box means you should **select all that apply**.
- If your answer is a string, make sure to put it in quotes. If your answer is a float, make sure to include a decimal point.
- You may use one handwritten sheet of notes. No calculators and no computers.

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By signing below, you are agreeing that you will behave honestly and fairly during and after this quiz.

Signature:

# Version A

Please do not open your quiz until instructed to do so.

## Question 1

You want to test the following hypotheses:

**Null Hypothesis:** Everyone who applies for an internship at Google has **a 20% chance** of receiving a job offer, independently of all other applicants.

**Alternative Hypothesis:** Everyone who applies for an internship at Google has **a more than 20% chance** of receiving a job offer, independently of all other applicants.

To test these hypotheses, you collected information from 50 applicants and found that 16 of them received a job offer.

**a)** Fill in the blanks in the code below to calculate the p-value for a hypothesis test where the test statistic is the number of applicants, out of 50, who receive a job offer.

```
offers_array = np.array([])

for i in np.arange(10000):
    num_offers = ___(a)___
    offers_array = ___(b)___

p_value = ___(c)___
p_value
```

(a):

(b):

(c):

**b)** Suppose the p-value comes out to 0.03. What conclusion do we draw?

◯ We reject the null hypothesis at both the 0.01 and 0.05 significance levels.

◯ We fail to reject the null hypothesis at the 0.01 significance level and we reject the null hypothesis at the 0.05 significance level.

◯ We reject the null hypothesis at the 0.01 significance level and we fail to reject the null hypothesis at the 0.05 significance level.

◯ We fail to reject the null hypothesis at both the 0.01 and 0.05 significance levels.

**c)** Which of the following test statistics would have also been appropriate to test these hypotheses? Select all that apply.

- ☐ Number of applicants, out of 50, who do not receive a job offer.
- ☐ Proportion of applicants that receive a job offer.
- ☐ The sum of the absolute differences between $[0.2, 0.8]$ and the distribution of applicants into two categories: those getting a job offer and those not getting a job offer.
- ☐ Absolute difference between 20 and the number of applicants, out of 50, who receive a job offer.
- ☐ None of the above.

## Question 2

According to Indeed, a popular job website, the hourly pay for data science interns across the US has a mean of 24 and a standard deviation of 6. You take a random sample of 64 data science interns. In your sample, the hourly pay has a mean of 25 and a standard deviation of 4. Suppose you bootstrap your sample 10,000 times, calculate the mean hourly pay from each resample, and plot a histogram of these resampled means. Which of the following best describes this histogram?

- ◯ Roughly normal with a mean of 24 and a standard deviation of 6.
- ◯ Roughly normal with a mean of 24 and a standard deviation of 0.75.
- ◯ Roughly normal with a mean of 25 and a standard deviation of 4.
- ◯ Roughly normal with a mean of 25 and a standard deviation of 0.5.
- ◯ None of the above.

## Question 3

You are interested in estimating the average wait time between an interview and an internship offer being made. You take a random sample of $n$ internship offers and find that in this sample, the average wait time is $d$ days and the standard deviation is 4 days.

You construct a 95% CLT-based confidence interval for the true average wait time, in days, which comes out to $[10.4, 13.6]$. Find $n$ and $d$.

$n =$ ☐ $\quad\quad$ $d =$ ☐

**Reminders:**

- **Write your PID** on the front page and on the top right corner of page 3.
- If your answer is a string, make sure to put it in **quotes**.
- If your answer is a float, make sure to include a **decimal point**.
- Stay in your seat until the quiz is over. No leaving early.